

MPhil Thesis: Empirical Results

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1 RQ1: Overall Innovation Response

1.1 Patents

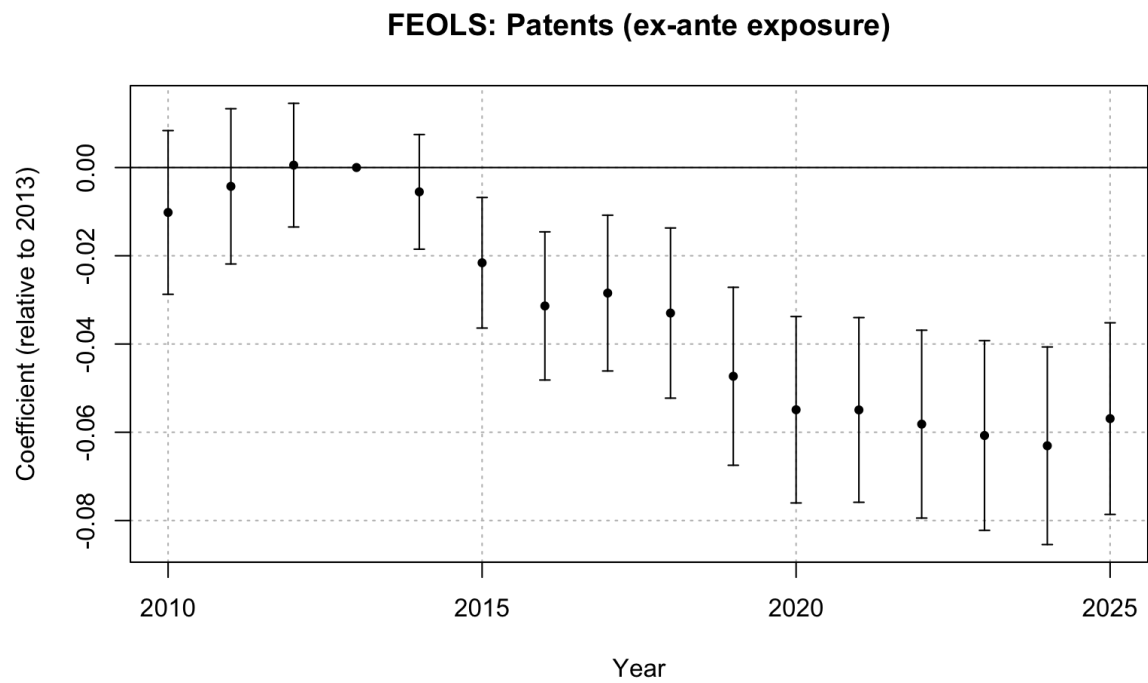


Figure 1: FEOLS event study: patents, ex-ante Medicaid exposure

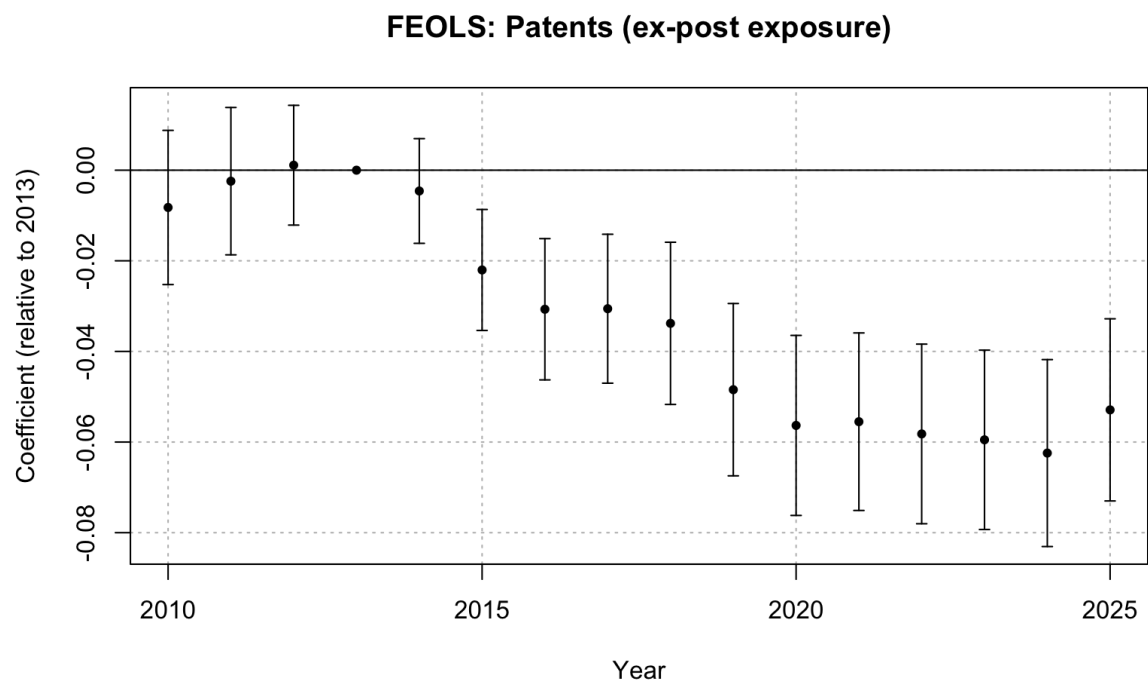


Figure 2: FEOLS event study: patents, ex-post Medicaid exposure

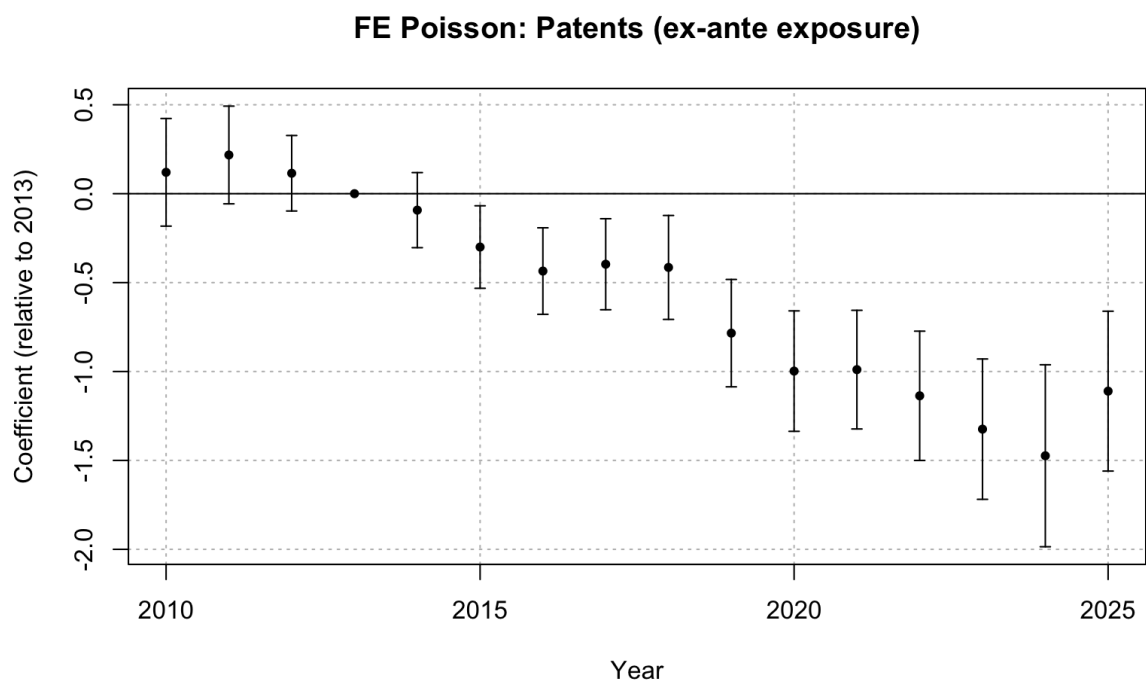


Figure 3: FE Poisson event study: patents, ex-ante Medicaid exposure

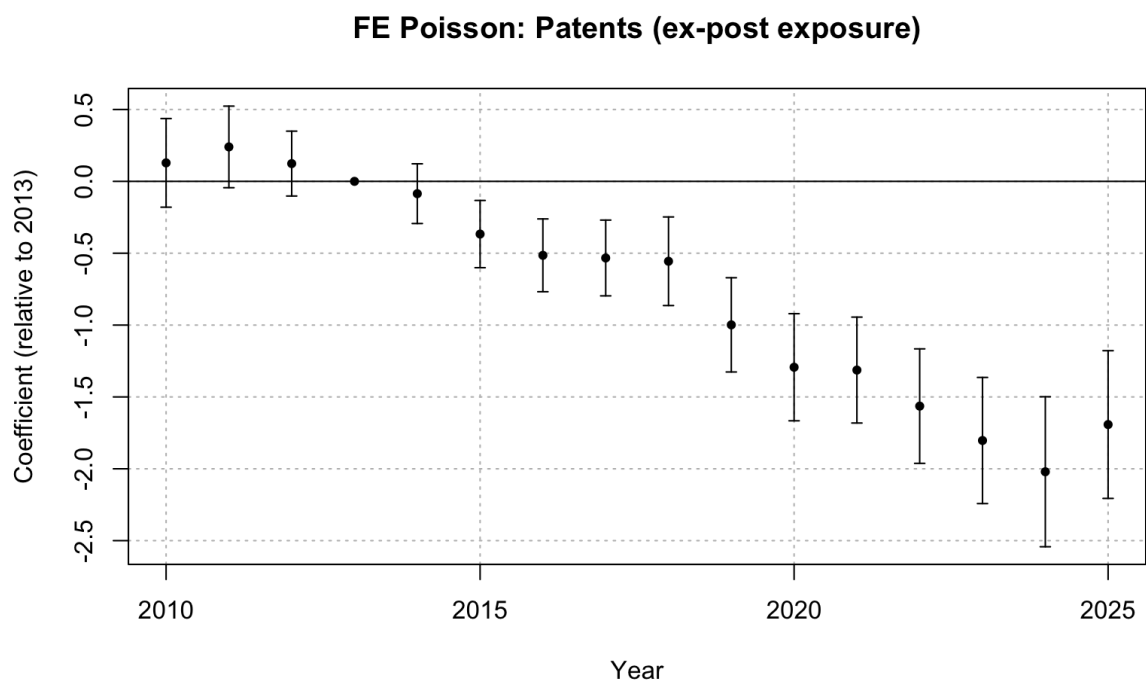


Figure 4: FE Poisson event study: patents, ex-post Medicaid exposure

	FEOLS: Ex Ante	FEOLS: Ex Post	FEPOIS: Ex Ante	FEPOIS: Ex Post
2010 $\times$ Medicaid Share	−0.010 (0.009)	−0.008 (0.009)	0.120 (0.154)	0.129 (0.157)
2011 $\times$ Medicaid Share	−0.004 (0.009)	−0.002 (0.008)	0.218 (0.140)	0.240 (0.145)
2012 $\times$ Medicaid Share	0.001 (0.007)	0.001 (0.007)	0.115 (0.108)	0.124 (0.115)
2014 $\times$ Medicaid Share	−0.006 (0.007)	−0.005 (0.006)	−0.092 (0.108)	−0.086 (0.106)
2015 $\times$ Medicaid Share	−0.022** (0.008)	−0.022** (0.007)	−0.300* (0.118)	−0.367** (0.119)
2016 $\times$ Medicaid Share	−0.031*** (0.009)	−0.031*** (0.008)	−0.435*** (0.124)	−0.515*** (0.129)
2017 $\times$ Medicaid Share	−0.028** (0.009)	−0.031*** (0.008)	−0.397** (0.131)	−0.533*** (0.134)
2018 $\times$ Medicaid Share	−0.033*** (0.010)	−0.034*** (0.009)	−0.415** (0.149)	−0.556*** (0.157)
2019 $\times$ Medicaid Share	−0.047*** (0.010)	−0.048*** (0.010)	−0.784*** (0.154)	−0.998*** (0.167)
2020 $\times$ Medicaid Share	−0.055*** (0.011)	−0.056*** (0.010)	−0.998*** (0.173)	−1.293*** (0.190)
2021 $\times$ Medicaid Share	−0.055*** (0.011)	−0.056*** (0.010)	−0.989*** (0.170)	−1.313*** (0.188)
2022 $\times$ Medicaid Share	−0.058*** (0.011)	−0.058*** (0.010)	−1.137*** (0.186)	−1.564*** (0.203)
2023 $\times$ Medicaid Share	−0.061*** (0.011)	−0.060*** (0.010)	−1.324*** (0.201)	−1.803*** (0.224)
2024 $\times$ Medicaid Share	−0.063*** (0.011)	−0.062*** (0.011)	−1.474*** (0.261)	−2.021*** (0.266)
2025 $\times$ Medicaid Share	−0.057*** (0.011)	−0.053*** (0.010)	−1.110*** (0.229)	−1.692*** (0.262)
Num. obs.	21 126 784	21 126 784	5 203 776	5 203 776
R2	0.392	0.392		

* p < 0.05, ** p < 0.01, *** p < 0.001

1.2 FDA Approvals

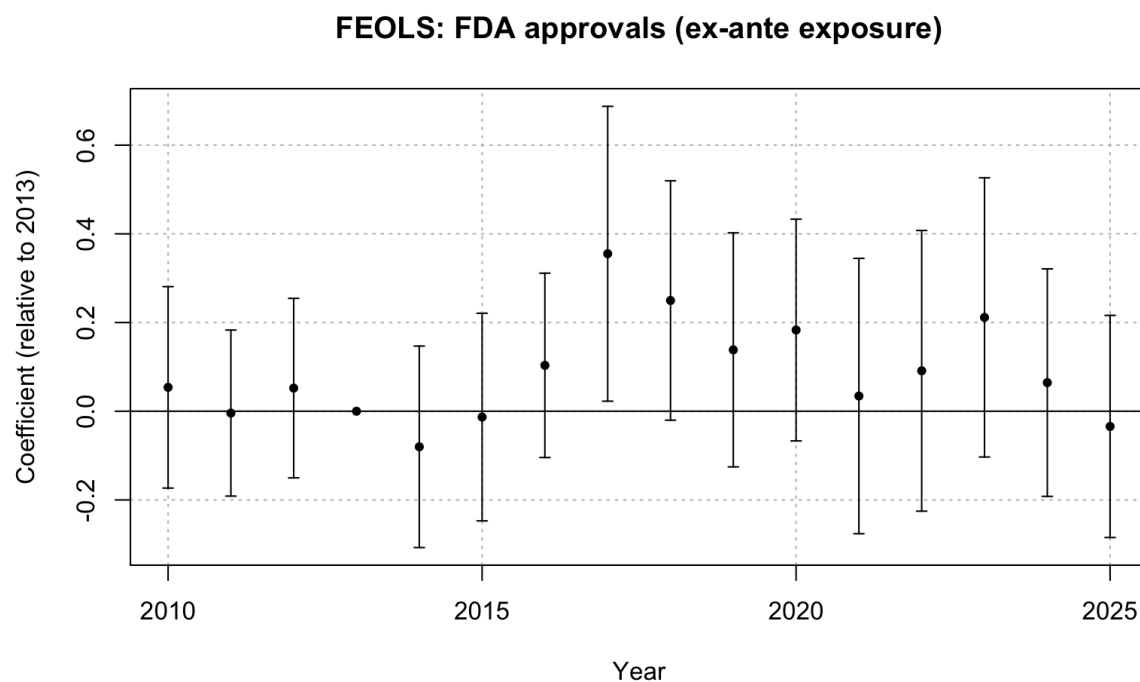


Figure 5: FEOLS event study: FDA approvals, ex-ante Medicaid exposure

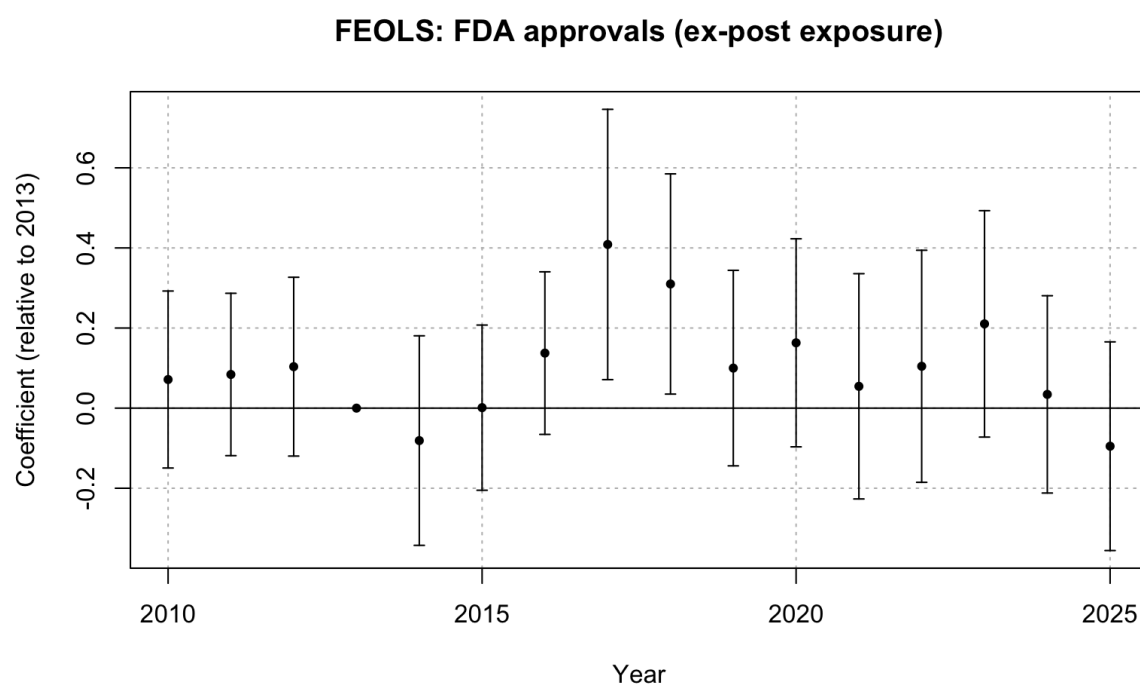


Figure 6: FEOLS event study: FDA approvals, ex-post Medicaid exposure

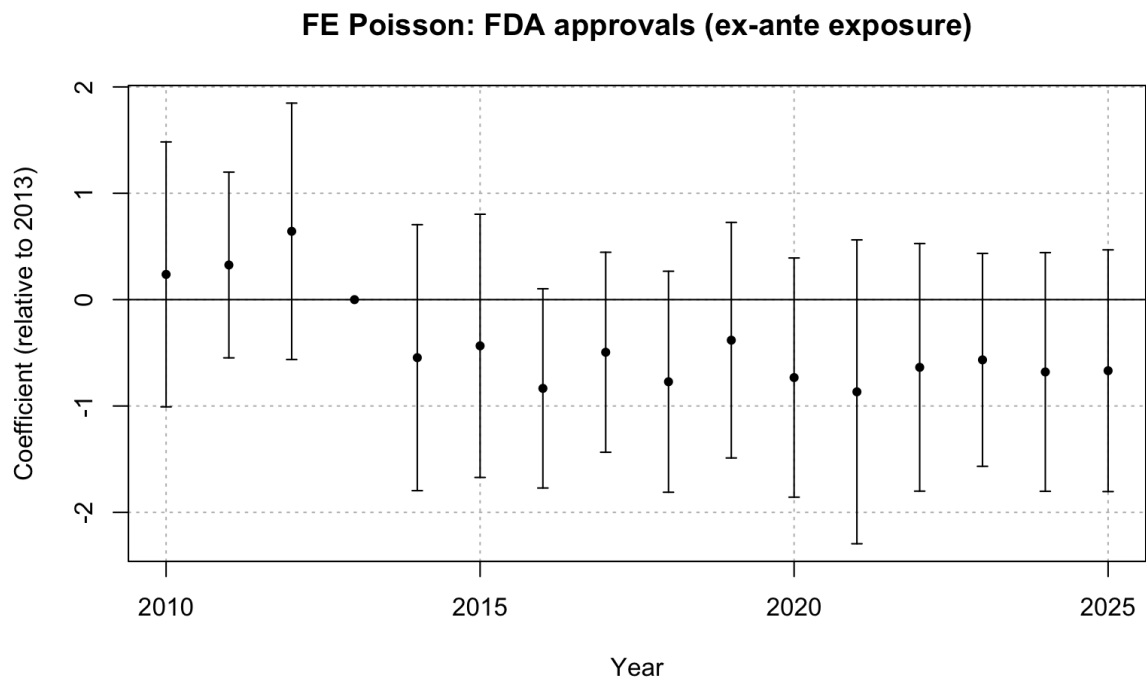


Figure 7: FE Poisson event study: FDA approvals, ex-ante Medicaid exposure

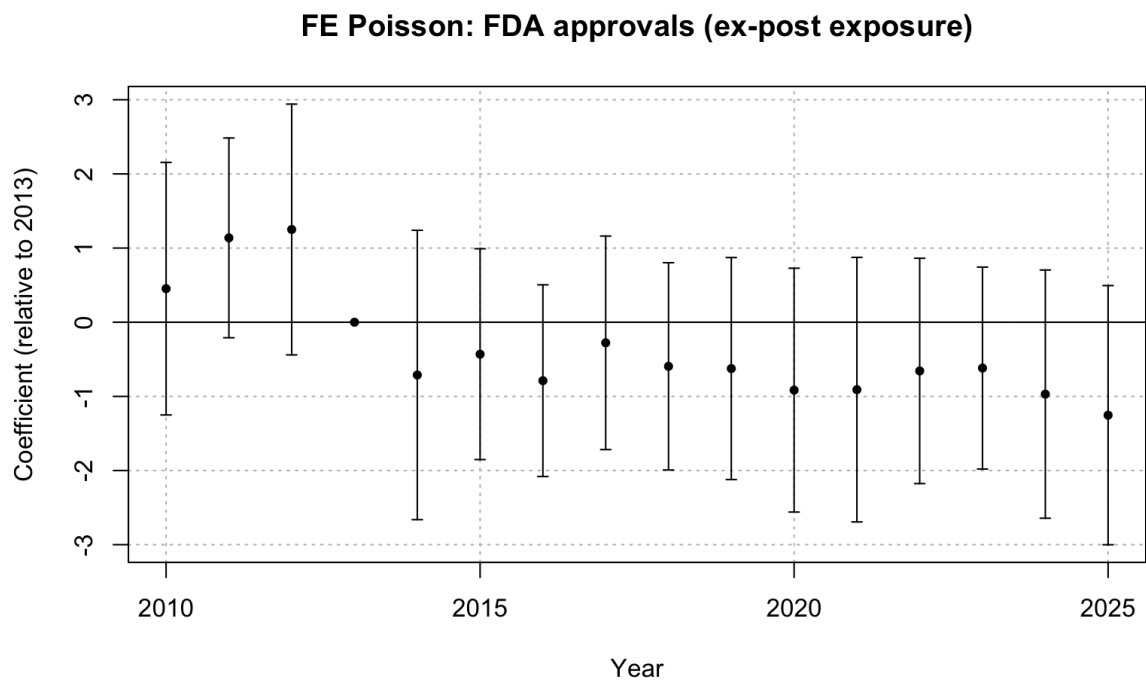


Figure 8: FE Poisson event study: FDA approvals, ex-post Medicaid exposure

	FEOLS: Ex Ante	FEOLS: Ex Post	FEPOIS: Ex Ante	FEPOIS: Ex Post
2010 × Medicaid Share	0.054 (0.116)	0.071 (0.113)	0.237 (0.636)	0.453 (0.869)
2011 × Medicaid Share	−0.004 (0.095)	0.084 (0.103)	0.326 (0.446)	1.138 (0.687)
2012 × Medicaid Share	0.052 (0.103)	0.104 (0.114)	0.643 (0.615)	1.250 (0.863)
2014 × Medicaid Share	−0.080 (0.116)	−0.081 (0.133)	−0.545 (0.638)	−0.711 (0.995)
2015 × Medicaid Share	−0.013 (0.119)	0.001 (0.105)	−0.434 (0.631)	−0.431 (0.725)
2016 × Medicaid Share	0.103 (0.106)	0.137 (0.103)	−0.834 (0.478)	−0.788 (0.659)
2017 × Medicaid Share	0.355* (0.169)	0.409* (0.172)	−0.495 (0.480)	−0.278 (0.735)
2018 × Medicaid Share	0.250 (0.137)	0.310* (0.140)	−0.772 (0.530)	−0.595 (0.713)
2019 × Medicaid Share	0.138 (0.134)	0.100 (0.124)	−0.381 (0.565)	−0.625 (0.764)
2020 × Medicaid Share	0.183 (0.127)	0.163 (0.132)	−0.733 (0.574)	−0.916 (0.839)
2021 × Medicaid Share	0.034 (0.158)	0.054 (0.143)	−0.867 (0.729)	−0.909 (0.910)
2022 × Medicaid Share	0.091 (0.161)	0.104 (0.148)	−0.637 (0.594)	−0.657 (0.775)
2023 × Medicaid Share	0.212 (0.160)	0.210 (0.144)	−0.566 (0.511)	−0.618 (0.695)
2024 × Medicaid Share	0.065 (0.131)	0.034 (0.125)	−0.680 (0.573)	−0.969 (0.854)
2025 × Medicaid Share	−0.034 (0.128)	−0.095 (0.133)	−0.668 (0.580)	−1.254 (0.892)
Num. obs.	604 800	604 800	168 704	168 704
R2	0.314	0.314		

* p < 0.05, ** p < 0.01, *** p < 0.001

2 RQ2: Medicaid Demand and PDL Exposure

2.1 Patents

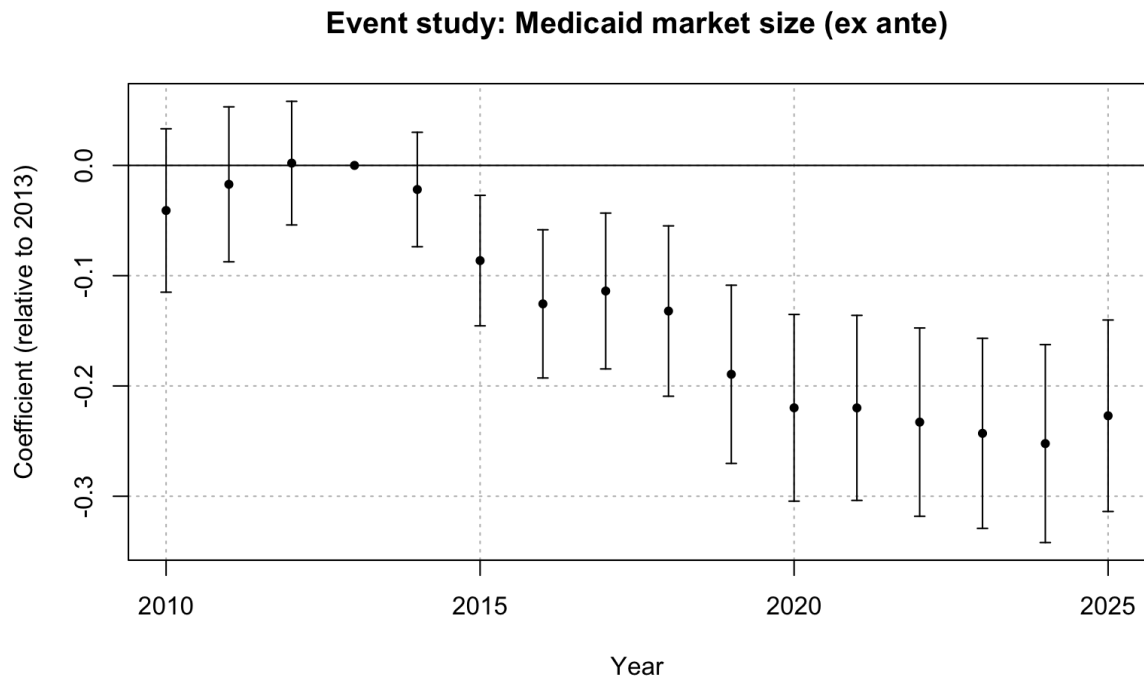


Figure 9: FEOLS event study: patents, Medicaid exposure (ex-ante)

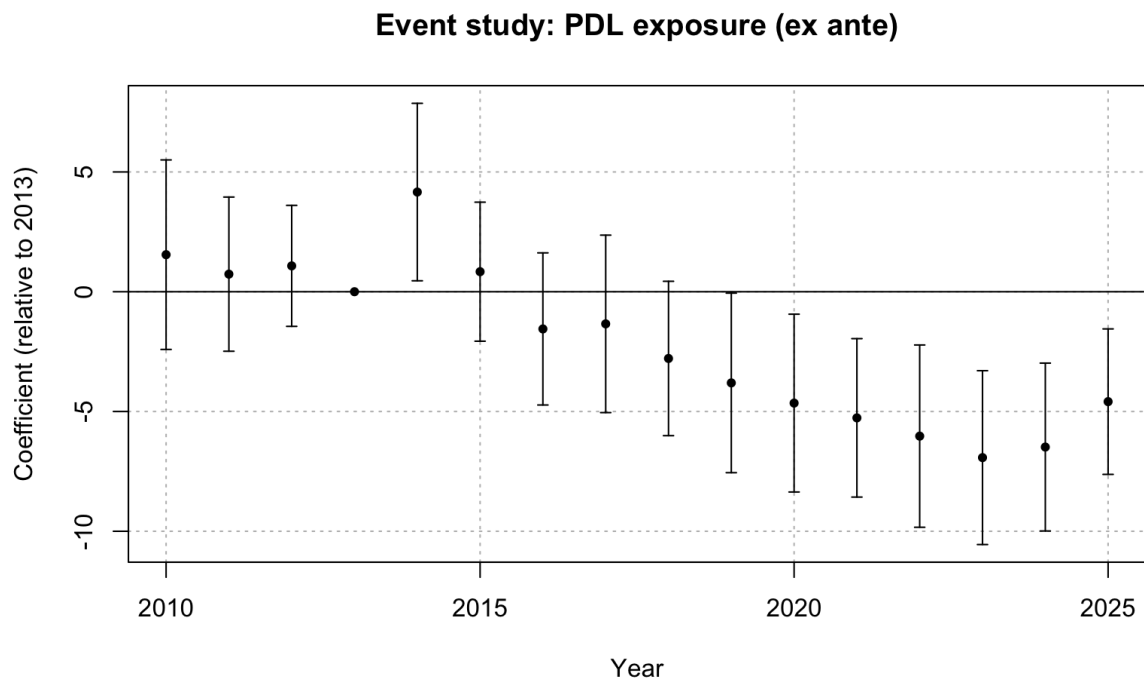


Figure 10: FEOLS event study: patents, PDL exposure (ex-ante)

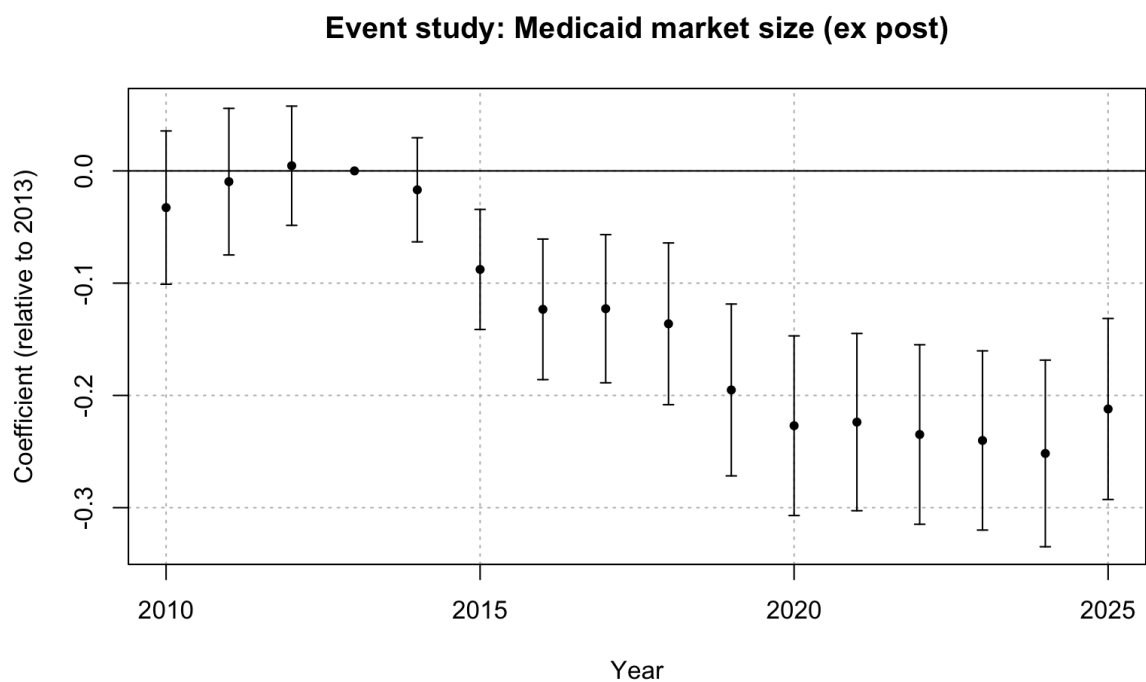


Figure 11: FEOLS event study: patents, Medicaid exposure (ex-post)

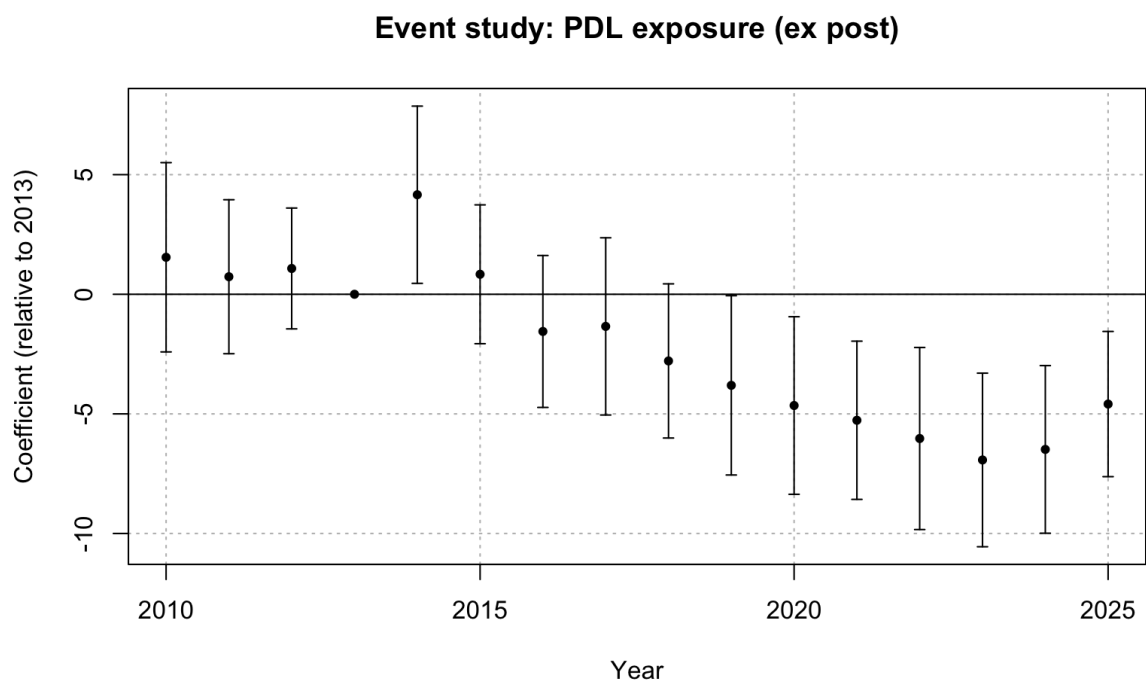


Figure 12: FEOLS event study: patents, PDL exposure (ex-post)

	FEOLS: Ex Ante		FEOLS: Ex Post	
	Medicaid Market Share	PDL Exposure	Medicaid Market Share	PDL Exposure
2010	−0.041 (0.038)	1.546 (2.018)	−0.033 (0.035)	1.546 (2.018)
2011	−0.017 (0.036)	0.734 (1.642)	−0.010 (0.033)	0.734 (1.642)
2012	0.002 (0.029)	1.079 (1.289)	0.005 (0.027)	1.079 (1.289)
2014	−0.022 (0.026)	4.161* (1.890)	−0.017 (0.024)	4.161* (1.890)
2015	−0.086** (0.030)	0.836 (1.480)	−0.088** (0.027)	0.836 (1.480)
2016	−0.126*** (0.034)	−1.554 (1.620)	−0.123*** (0.032)	−1.554 (1.620)
2017	−0.114** (0.036)	−1.342 (1.890)	−0.123*** (0.034)	−1.342 (1.890)
2018	−0.132*** (0.039)	−2.786 (1.644)	−0.136*** (0.037)	−2.787 (1.644)
2019	−0.189*** (0.041)	−3.806* (1.912)	−0.195*** (0.039)	−3.807* (1.912)
2020	−0.220*** (0.043)	−4.649* (1.894)	−0.227*** (0.041)	−4.650* (1.894)
2021	−0.220*** (0.043)	−5.266** (1.687)	−0.224*** (0.040)	−5.267** (1.687)
2022	−0.233*** (0.044)	−6.029** (1.941)	−0.235*** (0.041)	−6.029** (1.941)
2023	−0.243*** (0.044)	−6.925*** (1.851)	−0.240*** (0.041)	−6.926*** (1.851)
2024	−0.252*** (0.046)	−6.484*** (1.788)	−0.252*** (0.042)	−6.485*** (1.788)
2025	−0.227*** (0.044)	−4.588** (1.549)	−0.212*** (0.041)	−4.589** (1.549)
Num. obs.	5 281 696		5 281 696	
R2	0.535		0.535	

* p < 0.05, ** p < 0.01, *** p < 0.001

2.2 FDA Approvals

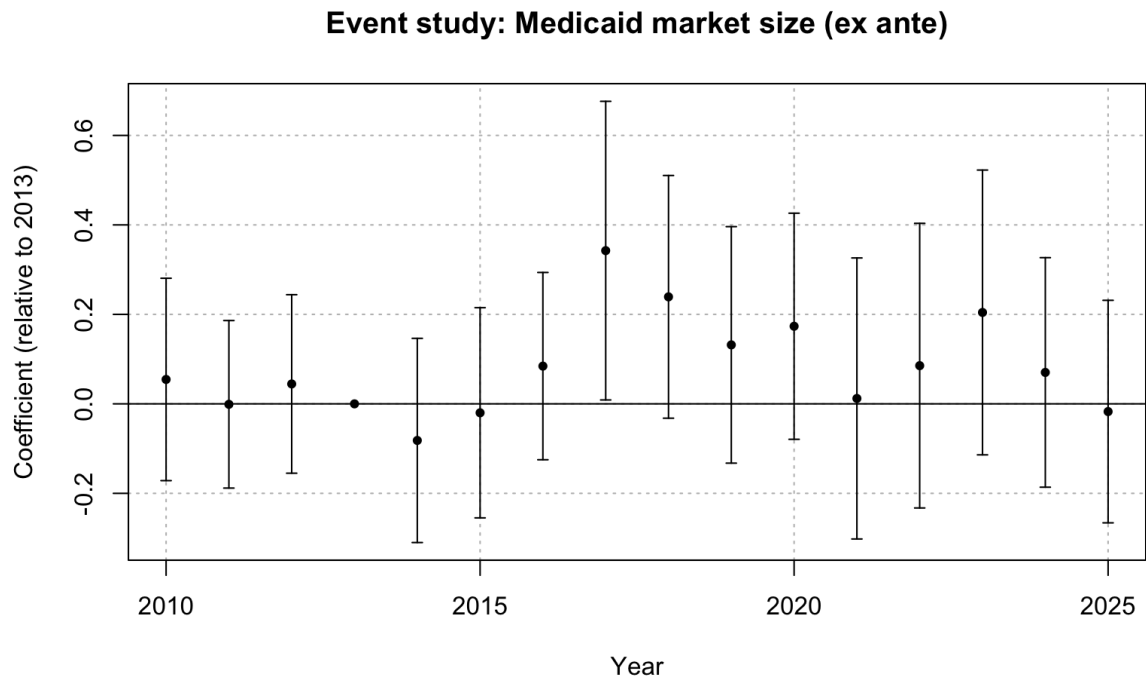


Figure 13: FEOLS event study: FDA approvals, Medicaid exposure (ex-ante)

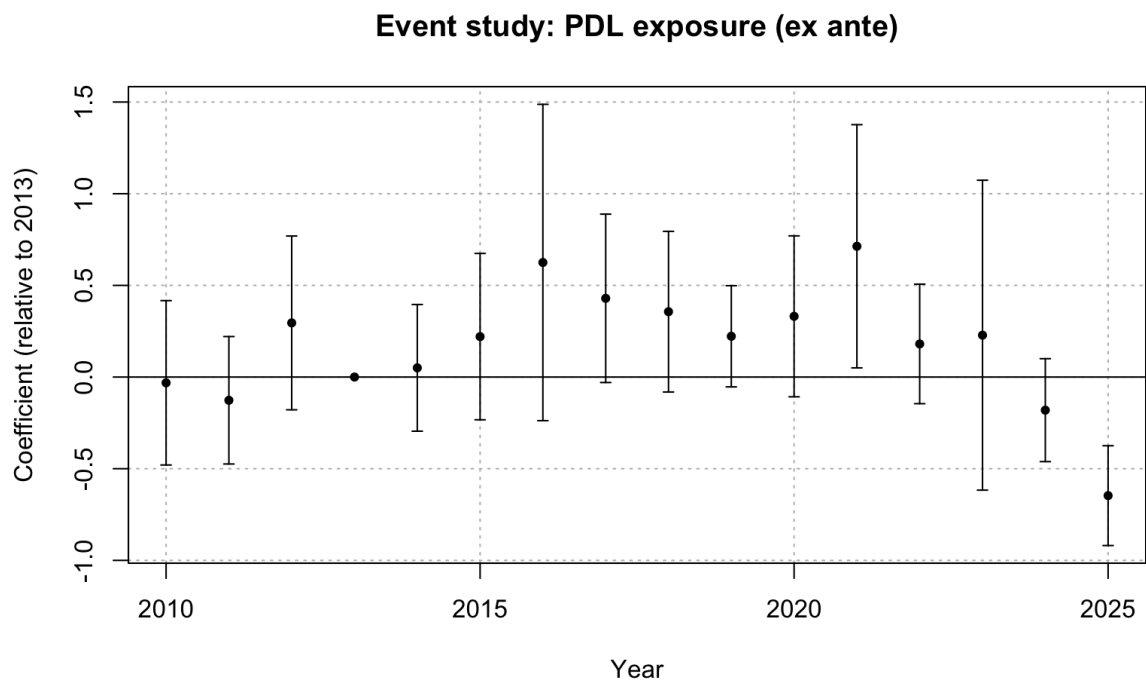


Figure 14: FEOLS event study: FDA approvals, PDL exposure (ex-ante)

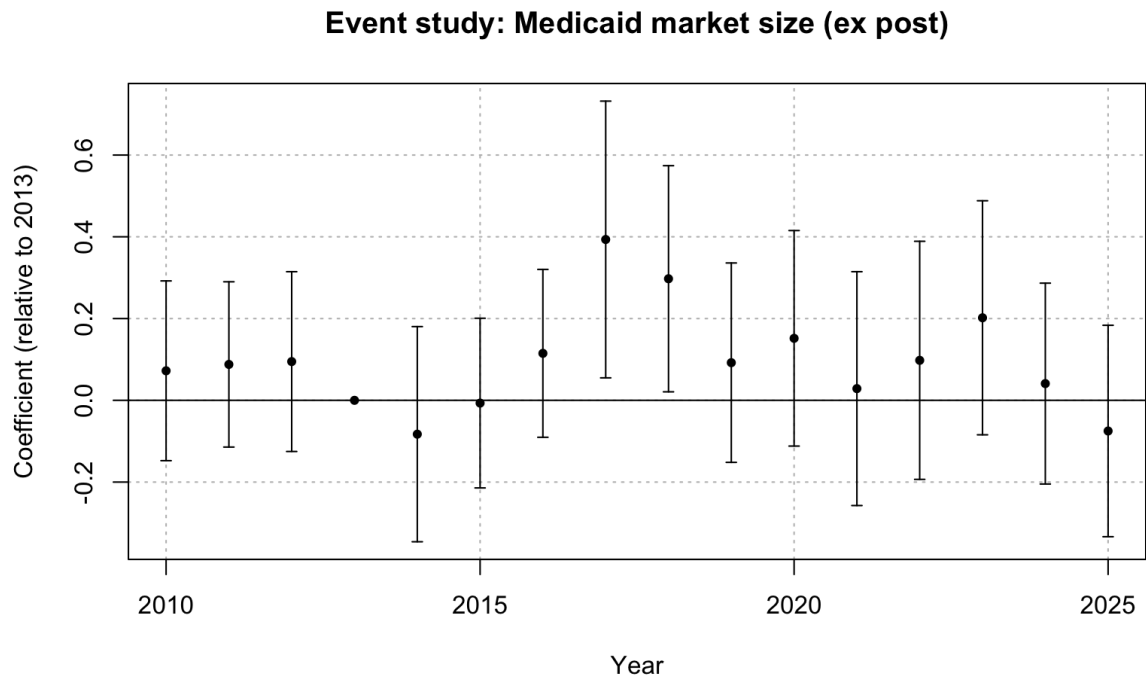


Figure 15: FEOLS event study: FDA approvals, Medicaid exposure (ex-post)

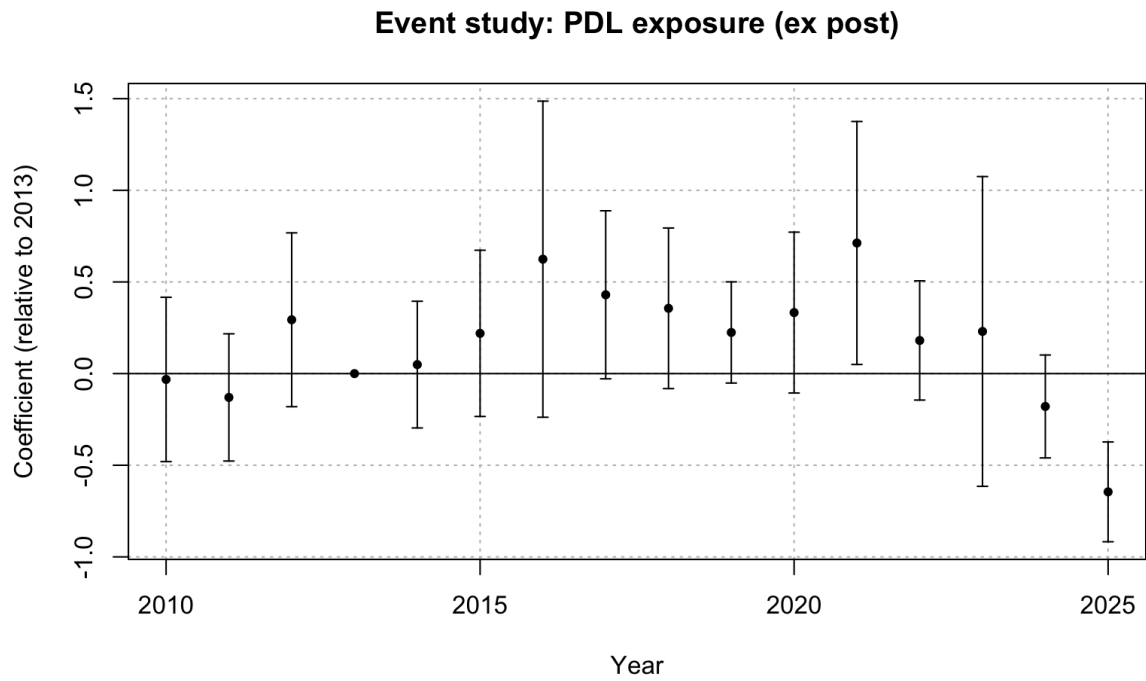


Figure 16: FEOLS event study: FDA approvals, PDL exposure (ex-post)

	FEOLS: Ex Ante		FEOLS: Ex Post	
	Medicaid Market Share	PDL Exposure	Medicaid Market Share	PDL Exposure
2010	0.055 (0.115)	−0.031 (0.228)	0.072 (0.112)	−0.032 (0.228)
2011	−0.001 (0.095)	−0.127 (0.177)	0.088 (0.103)	−0.130 (0.177)
2012	0.044 (0.102)	0.295 (0.241)	0.095 (0.112)	0.294 (0.241)
2014	−0.082 (0.116)	0.050 (0.176)	−0.083 (0.134)	0.049 (0.176)
2015	−0.020 (0.120)	0.221 (0.231)	−0.007 (0.106)	0.220 (0.231)
2016	0.084 (0.107)	0.625 (0.439)	0.115 (0.105)	0.624 (0.439)
2017	0.342* (0.170)	0.429 (0.234)	0.393* (0.172)	0.430 (0.233)
2018	0.239 (0.138)	0.356 (0.223)	0.297* (0.141)	0.356 (0.223)
2019	0.132 (0.135)	0.222 (0.141)	0.092 (0.124)	0.224 (0.141)
2020	0.173 (0.129)	0.331 (0.224)	0.152 (0.134)	0.333 (0.223)
2021	0.012 (0.160)	0.713* (0.338)	0.029 (0.146)	0.713* (0.338)
2022	0.085 (0.162)	0.181 (0.166)	0.098 (0.148)	0.181 (0.166)
2023	0.204 (0.162)	0.228 (0.430)	0.202 (0.146)	0.230 (0.430)
2024	0.070 (0.131)	−0.181 (0.143)	0.041 (0.125)	−0.179 (0.143)
2025	−0.017 (0.127)	−0.647*** (0.139)	−0.075 (0.132)	−0.645*** (0.139)
Num. obs.	604 800		604 800	
R2	0.315		0.315	

* p < 0.05, ** p < 0.01, *** p < 0.001

3 RQ3: Therapeutic-Class Heterogeneity

3.1 Patents

	FEOLS: Ex Ante	FEOLS: Ex Post
medicaid_market_share $\times$ post $\times$ atc_level1 = A	−0.165*** (0.033)	−0.157*** (0.031)
medicaid_market_share $\times$ post $\times$ atc_level1 = B	0.134*** (0.024)	0.151*** (0.027)
medicaid_market_share $\times$ post $\times$ atc_level1 = C	−0.123*** (0.019)	−0.094*** (0.015)
medicaid_market_share $\times$ post $\times$ atc_level1 = D	−0.030 (0.108)	−0.017 (0.062)
medicaid_market_share $\times$ post $\times$ atc_level1 = G	0.037 (0.056)	0.032 (0.049)
medicaid_market_share $\times$ post $\times$ atc_level1 = H	0.876*** (0.136)	0.619*** (0.096)
medicaid_market_share $\times$ post $\times$ atc_level1 = J	0.055 (0.054)	0.035 (0.034)
medicaid_market_share $\times$ post $\times$ atc_level1 = L	0.346 (0.452)	0.443 (0.578)
medicaid_market_share $\times$ post $\times$ atc_level1 = M	−0.336*** (0.076)	−0.212*** (0.048)
medicaid_market_share $\times$ post $\times$ atc_level1 = N	−0.125*** (0.028)	−0.174*** (0.039)
medicaid_market_share $\times$ post $\times$ atc_level1 = R	0.005 (0.003)	0.005 (0.003)
medicaid_market_share $\times$ post $\times$ atc_level1 = S	0.093*** (0.017)	0.105*** (0.020)
Num. obs.	19 617 728	19 617 728
R2	0.247	0.247

* p < 0.05, ** p < 0.01, *** p < 0.001

3.2 FDA Approvals

	FEOLS: Ex Ante	FEOLS: Ex Post
medicaid_market_share $\times$ post $\times$ atc_level1 = A	1.211** (0.391)	1.301** (0.420)
medicaid_market_share $\times$ post $\times$ atc_level1 = B	-0.396 (2.567)	-0.121 (0.784)
medicaid_market_share $\times$ post $\times$ atc_level1 = C	0.229 (0.171)	0.373 (0.278)
medicaid_market_share $\times$ post $\times$ atc_level1 = D	0.707** (0.270)	0.881** (0.337)
medicaid_market_share $\times$ post $\times$ atc_level1 = G	-2.516 (2.666)	-1.058 (1.121)
medicaid_market_share $\times$ post $\times$ atc_level1 = H	1.144 (1.358)	0.446 (0.530)
medicaid_market_share $\times$ post $\times$ atc_level1 = J	-0.145 (0.165)	-0.078 (0.089)
medicaid_market_share $\times$ post $\times$ atc_level1 = L	-0.321 (2.789)	-0.244 (2.115)
medicaid_market_share $\times$ post $\times$ atc_level1 = M	0.022 (1.127)	0.017 (0.858)
medicaid_market_share $\times$ post $\times$ atc_level1 = N	0.034 (0.074)	0.047 (0.101)
medicaid_market_share $\times$ post $\times$ atc_level1 = R	0.298 (0.217)	0.178 (0.130)
medicaid_market_share $\times$ post $\times$ atc_level1 = S	8.052* (3.737)	6.473* (3.004)
Num. obs.	561 600	561 600
R2	0.235	0.235

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

4 PDL Counterfactual Analysis

	Patents	FDA Approvals
Post $\times$ P&T Pass (Clinical Only)	-11.399* (4.663)	1.002 (0.633)
Post $\times$ Price Pass (Price Only)	-7.386*** (2.173)	0.057 (0.881)
Post $\times$ In PDL (Both Stages)	15.549** (5.215)	-0.112 (1.174)
Num. obs.	5 281 696	151 200
R2	0.535	0.609

* p < 0.05, ** p < 0.01, *** p < 0.001

References